

Long Questions

Q1: Consider a game of football with the following payoffs:

Q2: A common issue in our country is that households dump their waste on street

Q3: Every year, millions of Pakistanis attain legal working age of 16,

Q4: Consider a pay off matrix of Prisoners Dilemma, given as under -

Q5: State and prove the condition for Pareto optimum in exchange.

Q6: Describe the first & Second fundamental Law of Welfare Economics.

Q7: Discuss Nash Equilibrium in respect of Prisoners Dilemma.

Q8: Explain Global Risk Aversion & measure of relative risk aversion

Q9: Attributes of Public Goods.

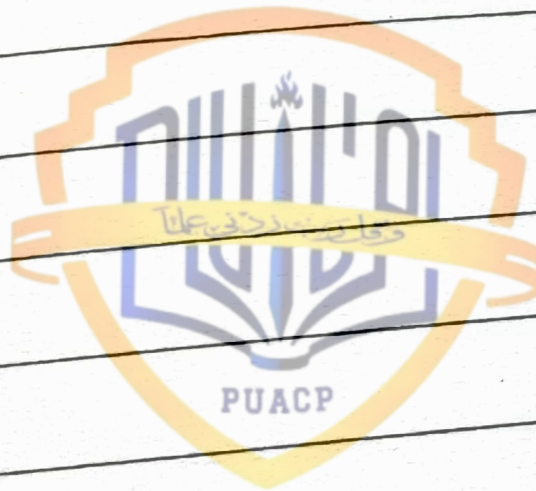
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Q10: Why does price leadership sometimes evolve in oligopolistic markets? — — —

Q11: Discuss the inefficiency of competition with externalities.

Q12: Explain the Stackelberg Model of duopoly.



Question #01

Answer:-

In a mixed strategy Nash Equilibrium, each player randomizes their strategy to keep their opponent indifferent between their available options.

Working

Steps To Determine Mixed-Strategy

Nash Equilibrium

1. **Identify Payoffs:** Consider the choices and payoffs for both players:

- **Kicker's Payoff:**

- If the goalie saves "Left" and the kicker kicks "Left", the kicker is stopped (Payoff = 0).

- If the goalie saves "right", the kicker scores (Payoff = 1).

- If the goalie saves "right" and the kicker kicks "right", the kicker

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is stopped (Payoff = 0).

- If the goalie saves "left" and the kicker kicks "right", the kicker scores (Payoff = 1)

• Goalie's Payoff:

- If the Goalie saves "left" and the kicker kicks "left", the goalie is successful (Payoff = 1).
- If the goalie saves "right" and the kicker kicks "left", the goalie fails (Payoff = 0).
- If the goalie saves "right" and the kicker kicks "right", the goalie is successful (Payoff = 1).
- If the goalie saves "left" and the kicker kicks "right", the goalie fails. (Payoff = 0).

Calculations

Calculate Expected Payoffs:

We'll say the kicker kicks left

probability π_K and the "right" with $1 - \pi_K$, and the goalie saves "left" with π_G and "right" with $1 - \pi_G$.

Goalie Expected Payoff

Set $1 - \pi_K$ equal to π_K ,

Solving for π_K :

$$1 - \pi_K = \pi_K$$

$$1 = \pi_K + \pi_K$$

$$1 = 2\pi_K$$

$$\boxed{\pi_K = 0.5}$$

Kicker Expected Payoff

Set π_G equals to $1 - \pi_G$,

Solving for π_G :

$$\pi_G = 1 - \pi_G$$

$$2\pi_G = 1$$

$$\boxed{\pi_G = 0.5}$$

Putting it Together In a Table:

Here is a mixed strategy Nash-Equilibrium for both players:

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Player	Strategy	Probability
Goalie	Save Left	0.5 (50%)
Goalie	Save Right	0.5 (50%)
Kicker	Kick Left	0.5 (50%)
Kicker	Kick Right	0.5 (50%)

Conclusion:

In this equilibrium:

- The goalie randomizes between saving left and saving right, each with a 50% probability.
- The kicker randomizes between kicking left and kicking right, each with a 50% probability.

Overview

Any deviations from these probabilities would increase the expected payoffs for the opponent, thus keeping both players indifferent and leading to a stable equilibrium.

Question #02 (Answers)

(a) What is the negative externalities?

Negative externalities are unintended and harmful side effects experienced by third parties due to an economic activity.

In this case, the negative externalities occurs due to dumping waste on the street includes:

- **Public Health Risk:** Accumulation of waste can lead to pollution of the environment, attract pests, spread diseases and contaminate water resources, affecting the health of community.
- **Environmental Damage:** Improper waste disposal can lead to pollution of the environment, harming wildlife and natural ecosystems.
- **Visuals & Olfactory Nuisance:** The

sight and smell of waste on the streets decrease the quality of life for residents and can deter tourism.

- **Economic Costs:** Increase public spending on cleaning and waste management, as well as possible decreases in property values.

(b)

How can govt. can regulate this externality by taxing?

The govt can regulate this externalities through a **Pigovian Tax**, which is designed to correct the negative external effects of a market activity:

- **Waste Disposal Tax:** Household could be charged a tax based on the amount of waste they generate. This provides an economic incentive to reduce waste production.

- and dispose of waste properly.
- **Levy On Illegal Dumping:**

Imposing fine on illegal dumping activities can deter households from disposing off waste irresponsibly.

(C)

How can Govt. can regulate?

(Answer)

Allocating property rights is a method through which the govt can transfer responsibility of maintaining cleanliness to individuals or communities.

Some possible ways include:

- **Assigning Property Rights:**

Individual households or communities are given ownership and maintenance responsibility for specific areas. This creates an

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incentive to keep the area clean: because they directly bear the consequences of pollution.

• Private Waste Management:

The govt could contract private firms for waste collection and disposal services, effectively transferring the responsibility to a company with the expertise and resources to handle it efficiently.

Conclusion

Implementing taxes and allocating property rights are both effective ways to regulate the negative externalities associated with waste dumping, aiming to encourage better waste management practices and improve the overall well-being of society.

Question #03

(Answer)

(a)

Asymmetric Information & Education Signaling In The Labor Market

How this is a problem of
asymmetric info for the employer:

Asymmetric Information occurs
when one party in a transaction
has more or better information
than the other party.

In labor market, this can
lead to significant challenges for
employers:

Adverse Selection: Employers cannot
easily distinguish between high
quality (good) and low quality
(bad) workers due to insufficient
information during the hiring
process. This lack of transparency
can lead to the hiring of

less competent employees, thereby reducing overall productivity.

- **Moral Hazards:** Once hired (Bad) workers might not perform their duties efficiently, knowing that their true performance is difficult to monitor and evaluate. This can lead undermine and employer's capacity to maintain a productive workforce.

- **Hiring Costs:** Employer might spend considerable resources on interviews, background checks and training programs to gather more information about applicants. This need for these measures increases operational costs.

(b)

How can college / University education -- -- ?

• Skill Verification:

Completing a college / University education requires students to overcome various academic challenges, demonstrating their dedication, intelligence and ability to learn. This signals to employers that these individuals are likely more capable than those without such credentials.

• Standardization:

Educational Qualifications help standardize the selection process. For instance, employers can change/compare applicants based on their GPA, coursework or reputation of the educational institutions, providing a more consistent basis for their hiring decisions.

• Reduce Hiring Risk:

By relying on educational signals, employers can reduce the risk of hiring unsuitable employees. This helps to mitigate issues related to adverse selection and moral hazard, leading to a more efficient and productive workforce.

Overview

Asymmetric information poses significant challenges in the labor market.

College or university education acts as a signaling mechanism, providing employers with valuable information about a candidate's skill and reliability. This reduces the risk and cost associated with hiring, improving overall market efficiency.

Q#04
(Ans)

Considering a Payoff Matrix ----

I see you were given a payoff matrix for the classic prisoner's Dilemma scenario involving two prisoners, A & B. Here is a recap of the setup:

	Prisoner (A) Confess	Prisoner (A) not Confess
Prisoner (B) Confess	A: 3 Ys; B: 3 Ys	A: 6 Months; B: 10 Ys
Prisoner (B) not Confess	A: 10 Ys; B: 6 Months	A: 2 Ys; B: 2 Ys

Let's discuss the Nash Equilibrium and Dominant Strategy in this scenario:

Nash Equilibrium

As discussed before in Short Question, Include the concept of Prisoner's Dilemma as well in this Long Question.

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Dominant Strategy

It is a strategy that is best for a player, regardless of what the opponent does.

Here, "Confessing" is a dominant strategy for both prisoners because:

For Prisoner A: Confessing results in either 3 years or 6 months, both better than the 2 years or 10 years outcomes when they don't confess.

For Prisoner B: Confessing results in either 3 years or 6 months, both better than the 2 years or 10 years outcomes when they don't confess.

⇒ Notice how if both prisoners confessing (3 years each) is not the optimal outcome collectively (both not confessing results in 2 years each) which highlight the dilemma aspect of this game.

O#05 (Answers)

Conditions For Pareto Optimum In Exchange

A Pareto Optimum (or Pareto efficient allocation) in exchange is a solution where no individual can be made better off without making someone else "worse off".

Conditions

(1) Individual Preferences:

Each person has their own preferences represented by indifference curves. These curves illustrate combinations of two goods that provide the same level of satisfaction or utility.

(2) Feasible Allocation:

The total allocation of goods

must be within the limits of what is available in the economy. This ensures that the sum of goods being distributed does not exceed the total amount available.

(3) Mutual Benefit From Trade

There should be potential for mutually beneficial trade. Both individuals should be able to improve their satisfaction by exchanging goods.

(4) Marginal Rate of Substitution

- The Marginal Rate of Substitution (MRS) is the rate at which one individual is willing to trade one good for another while maintaining the same level of satisfaction.

- For a Pareto Optimum, the MRS b/w two goods must be

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equal for all individuals involved.

This means both parties value the trade off between goods in the same way.

Mathematically:

$$MRS_A^{x,y} = MRS_B^{x,y}$$

where:

→ $MRS_A^{x,y}$: Marginal Rate of Substitution of Good "X" for Good "Y" for Person "A".

→ $MRS_B^{x,y}$: Marginal Rate of Substitution of Good "X" for Good "Y" for person "B".

Example:

(1) Person A: A's Utility Function: $U_A(x, y)$

The MRS for Person A is:

$$MRS_A^{x,y} = - \frac{\partial U_A / \partial x}{\partial U_A / \partial y}$$

(2) Person B: B's Utility function: $U_B(x, y)$

$$MRS_B^{x,y} = - \frac{\partial U_B / \partial x}{\partial U_B / \partial y}$$

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Conditions For Optimality

For the exchange to be Pareto optimal, the trade-offs or the rates at which both parties are willing to substitute goods (Good X for Good Y) must be equal.

As:

$$\frac{-\partial U_A / \partial X}{\partial U_A / \partial Y} = \frac{-\partial U_B / \partial X}{\partial U_B / \partial Y}$$

This ensures that both the parties are maximizing their satisfaction and no further mutually beneficial trades can be made.

Q#6

First & Second Fundamental Law of welfare Economics.

(Answers)

Let's delve into the two fundamental theorems of welfare economics, which form the bedrock of modern economics theory by explaining the efficient allocation of resources in a competitive model.

First Fundamental Theorem (The Efficiency Theorem)

Statement:

Every competitive equilibrium, or Walrasian equilibrium, leads to a Pareto efficient allocation of resources.

EXPLANATION:

- In a perfectly competitive

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market where all agents are price takers, and there are no externalities, public goods, or distortion like taxes or subsidies.

• This means that no individual in the economy can be made better off without making someone else worse off. This market ensures that the resources are allocated in the most efficient manner.

PROOF

Assumptions

- Markets are perfectly competitive
- No Externalities
- Complete Information
- Rational Behavior of agents

Mechanism

Key:

- Prices adjust to equate supply and demand.
- Consumers maximize utility given their budget constraints.
- Producers maximize profits given their cost structures.

Outcome

The choices of individuals and firms driven by maximizing behavior, align with market prices to ensure efficient resources allocation.

Second Fundamental Theorem (The Equity Theorem)

Statement:

Any Pareto efficient allocation can be achieved by a competitive equilibrium given an appropriate redistribution of initial endowments.

EXPLANATION:

This theorem separates concerns of efficiency (addressed by the first theorem) from equity. It states that it is possible to reach any desired Pareto efficient outcome by redistributing initial resources (wealth distribution) and then allowing market forces to act freely.

PROOF

Assumptions

- The same assumption as apply in Perfect competition.
- Society can redistribute initial endowments without causing market distortions.

Mechanism

- An appropriate redistribution of

- wealth or resources (e.g. through taxes, subsidies, or direct transfer) is implemented.
- After redistribution, the market operates freely without intervention.

Outcome

Post-redistribution, the competitive market equilibrium ensures the efficient allocation of resources as per the first theorem.

Summary

1st Fundamental Theorem:

Market equilibria are efficient.

Any efficient allocation can be attained with an appropriate redistribution of resources.

Overview

These theorems together highlight the power of markets in achieving efficiency and the role of redistributing in addressing equity without sacrificing efficiency.

Question #07

Nash Equil. In respect of Prisoner's
(Answer)

Nash Equilibrium In The Prisoner's Dilemma

The Prisoner's dilemma is a classic example in game theory that illustrates the concept of a Nash Equilibrium. In this scenario, two "Prisoners" must independently decide whether to confess to a crime or

remain silent. The outcome of each prisoner depends on the decision made by the others.

⇒ The Setup:

If both Confess: Each serves 3 years in prison.

If One Confess & Other Remain Silent: the confessor goes free while the silent one serves 10 years in prison.

If Both Remain Silent: each serves only 6 months.

Here's the payoff Matrix:

		Prisoner (B)	Prisoner (B)
		Confess	not Confess
Prisoner (A)	Confess	A: 3Ys; B: 3Ys	A: 0Ys; B: 10Ys
Prisoner (A)	not Confess	A: 10Ys; B: 0Ys	A: 6 months; B: 6 Months

Nash Equilibrium:

It occurs when each player chooses a strategy that maximizes their payoffs, given the strategy chosen by the other players.

In this dilemma, each prisoner has two options:

→ Confess or not Confess.

(1) Analyze Prisoner A's Choice

- If Prisoner B confesses:
 - Confess: Prisoner A gets 3 years
 - Not Confess: Prisoner A gets 10 years
- If Prisoner B does not Confess:
 - Confess: Prisoner A goes free (0 years)
 - Not Confess: Prisoner A gets 6 months

Analyze Prisoner B's Choice

- If Prisoner A confesses:
 - Confess: Prisoner B gets 3 years
 - Not Confess: Prisoner B gets 10 years

If Prisoner B does not Confess:

Confess: Prisoner B goes free (0 years)

Not Confess: Prisoner B gets (6 month).

Conclusion

Given these choices, both prisoners have a dominant strategy to confess because it results in a better outcome for themselves regardless of what the other prisoner decides.

Therefore;

If P-A confess, the best response for P-B is to confess (3 yrs vs 10 yrs)

If P-B Confesses, the best response for P-A is to confess (3 yrs vs 10 yrs).

Takeaway

The prisoner's dilemma describes that how individual rationality can lead to a suboptimal

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sub outcome for all the parties involved, highlighting the tension between individual incentives and collective welfare.

Q#08

Global Risk Aversion & Measures of Relative Risk Aversion.

(Answer)

Global Risk Aversion

It refers to the tendency of investors to avoid risk on a global scale, often in response to economic uncertainties or financial market volatility.

When the global risk aversion increases, investors tend to move their safer assets such as govt bonds, and away

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from riskier assets like stocks or emerging market investments. This behavior can lead to significant fluctuations in asset prices and capital flows across countries.

MEASURES

Measures Of Relative Risk Aversion

Relative risk aversion measures how risk-averse an individual is relative to changes in their wealth. It is often used to compare the risk preferences of individuals with different levels of wealth.

There are two main measures:

- (1) Arrow-Pratt Relative Risk Aversion
- (2) Constant Relative Risk Aversion

Arrow-Pratt Relative Risk Aversion (RRA)

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- The measure is calculated by multiplying the Arrow-Pratt Absolute Risk Aversion (ARA) coefficient by the level of wealth.

- The formula is:

$$RRA = W \cdot ARA$$

where "w" is the level of wealth and "ARA" is the Arrow-Pratt

- absolute Risk aversion coefficient.

Constant Relative Risk Aversion (CRRA)

- The measure indicates that an individual's degree of risk aversion remains constant regardless of changes in their wealth.
- For example; if an individual's utility function is $u(w) = -w^{-\gamma}$, the CRRA is equal to " γ ".
- The formula is:

$$RRA = \gamma$$

- whose " γ " is a constant representing the degree of relative risk aversion.

Overview

These measures help economists and financial analysts understand how individuals and markets respond to risk and market decisions about investment and asset allocation.

Q#69

Describe attributes of Public Goods.
(Answer)

Public Goods

Public goods are special types of goods or services that have unique characteristics distinguishing them from private goods.

Here are some key attributes:

Attributes Of Public Goods

(1) Non-Excludability

• Description:

It's difficult or impossible to exclude individuals from using the good once it has been provided. This means that everyone can access and benefit from the good without being prevented from doing so.

⇒ Example:

"National Defence" is non-excludable because once a country is protected, all citizens benefit from the protection regardless of whether they contribute to its funding.

(2) Non-Rivalry

Description:

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Once person's consumption of the good does not reduce the availability of the good for others. Multiple people can consume a good simultaneously without depleting it.

⇒ **Exemplifying:** 😊

Public parks are non-rivalrous because more than one person enjoying the park does not prevent others from enjoying it at the same time.

(3) Indivisibility

Description:

Public goods are often provided as a whole unit and cannot be divided into smaller parts that can be individually consumed.

⇒ **Example:**

A lighthouse serves as an

indivisible public good since it provides navigation aid to all ships in the vicinity.

(4) Collective Consumption

Description:

Street lighting provides illumination. Public goods are consumed collectively by the community, and their benefits are shared among all individuals within the community.

⇒ Example:

Street lighting provides illumination to all residents in a neighborhood collectively, rather than individually.

EXAMPLES:

Common examples of public goods are as follows:

- Clean Air: Once provided,

everyone can breathe it, and one person's use does not reduce its availability to others.

• Public Broadcasting:

Non-excludable and non-rivalrous as anyone can watch or listen to public radio or TV without restricting others.

• Public Health Initiatives:

Immunization campaigns reduce disease spread, benefiting everyone in the community collectively.

→ Challenges:

Challenges associated with Public Goods Are:

→ Free-Rider Problem

→ Provision & Financing

Overview

Understanding these attributes help

recognizing the importance of public goods and the role of govt. and institutions in ensuring their availability for the collective well-being.

Q#10

Why does price leadership sometime...
(Answer)

Price Leadership In Oligopolist Markets

Price Leadership

It is a phenomenon in oligopolist markets, where one firm (the price leader) sets the price for the product, and other firms in the market follow by setting their prices accordingly.

This behaviour can evolve due

to various reasons:

- 1) Market Dominants
- 2) Cost Efficiency
- 3) Reducing Uncertainty
- 4) Tacit Collusion (غیر واضح)

Market Dominants

A single firm may dominate the market in terms of size, production capacity, or brand reputation, enabling it to influence market prices.

Cost Efficiency

The price leader might have lower costs of production, allowing it to set a competitive price that other firms cannot undercut without incurring losses.

Reducing Uncertainty

In an oligopoly, firms are face uncertainty regarding

competitor's pricing strategy. Following a leader reduces the risk of price wars and ensure stable profits.

Tacit Collusion

Without explicit agreements, firms might follow a leader to implicitly cooperate, maximizing their joint profits while avoiding legal repercussions.

Determining Profit Maximizing Price

The price leader determines the profit-maximizing price by considering several factors:

Demand Curve Analysis

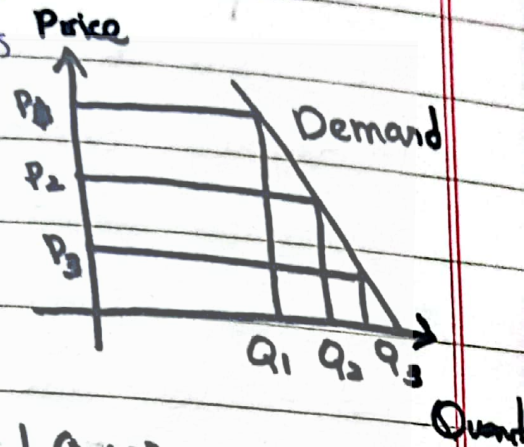
Marginal Cost (MC)

Marginal Revenue (MR)

Profit Maximization Rule

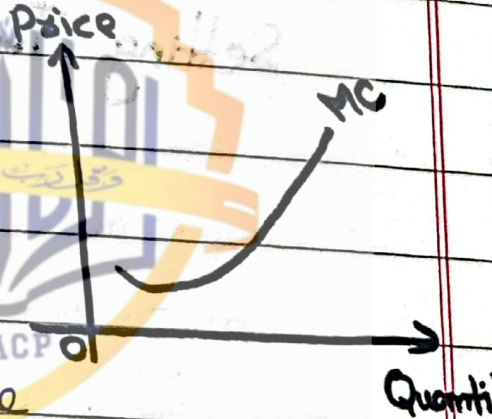
Demand Curve Analysis

The leader assesses the market demand for its product. The demand curve shows the relationship between price and Quantity demanded.



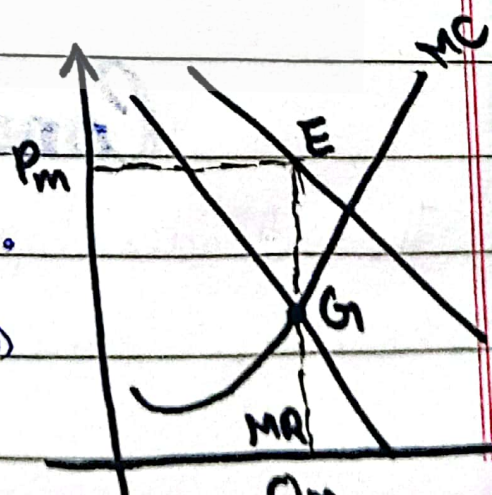
Marginal Cost (MC)

The cost of producing an additional unit of the product. The leader determines its MC to understand the cost implications of different output levels.



Marginal Revenue (MR)

The additional revenue generated from selling one more unit. The leader calculates



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MR to identify the revenue implications of different output levels.

Profit - Maximizing Rule

The leader sets its output where Marginal revenue (MR) equals Marginal Cost (MC). This ensures that producing any more, less will not increase profit.

$$MR = MC$$

Setting The Price

Once the profit-maximizing output is determined, the leader uses the market demand curve to set the corresponding price. The Quantity at which $MR = MC$ intersects the demand curve gives the price that maximizes profit.

Conclusion

The leadership in oligopolist market evolves due to market dominance.

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cost efficiency, uncertainty reduction, and tacit collusion. The price leader determines the profit-maximizing price by equating MR and MC and using the market demand curve to set the appropriate price.

Q#11

Inefficiency of competition with externalities.

(Answer)

Inefficiency Of Competition With Externalities

Externalities are costs or benefits experienced by third parties due to economic activities they are not directly involved in. There can be either positive (beneficial) effects,

or negative (harmful effects).

In competitive markets, externalities often lead to inefficiencies because market prices do not reflect the full social costs or benefits.

Negative Externalities:

Example: Pollution from a factory.

(1) Private Cost vs. Social Cost

- **Private Cost:** Costs incurred by the firm for production (e.g. labor, materials).
- **Social Cost:** Includes private costs and external costs (e.g. environmental damage, health impact).

(2) Market Outcomes:

- Firms produce where their private marginal cost (MC) equals the marginal benefits (MB) to consume.
- This leads to overproduction

because the firm does not amount account for the external costs imposed on society.

(3) Inefficiency:

Social Optimal production level is lower than the market production level.

The result is deadweight loss, a net loss of social welfare due to the divergence between private and social costs.

Positive Externalities

Example: Education

(1) Private & Social Benefits:

- **Private Benefits:** Benefits received by individuals (e.g.; higher earnings).
- **Social Benefits:** Include Private benefits and external benefits (e.g.; reduced crime rates, informed citizenry).

(2) Market Output:

- Individuals consume education where their private marginal benefits (MB) equals their marginal costs (MC).
- This leads to underconsumption because individuals do not consider the additional social benefits for their education.

(3) Inefficiency:

Social optimal consumption level is higher than the market consumption level.

The result is a deadweight loss: a net loss of social welfare due to divergence between price and social benefits.

Solution:

Here are some solutions to

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address inefficiencies:

- **Govt Interventions** by imposing taxes and subsidies.
- **Regulation and Standards** setting for the regulatory decisions.
- **Market-Based Solutions** Allowing firms for tradeable Permits and public Provision.

Overview

Competition with externalities leads to inefficiency because market prices do not reflect the full social costs or benefits of economic activities.

Q #12

Explain the Stackelberg Model of Duopoly.

(Answer)

Stacklberg Model Of Duopoly

The Stacklberg Model is a strategic game in economics that describes the behavior of firms in an oligopolist market, specifically a duopoly (two firms).

This model is named after the German economist Heinrich Von Stacklberg, who introduced it in 1934.

Key Features:

(1) Leader-Follower Structure:

- One firm (the leader) sets its output level first.
- The other firm (the follower) observes the leader's output and then decides its own output level.

(2) Sequential Decision-Making:

- The leader firm moves first

and the follower from reacts to the leader's decision.

- This sequential nature differentiate the Stackelberg model from the Cournot Model, whose firm decide simultaneously.

Steps

Steps In Stackelberg Model

(1) Leader's Decision:

- The leader firm determines its optimal output by considering how the followers will react to its decision.
- The leader maximizes its output by anticipating the followers response.

(2) Follower's Reaction Function:

- The followers observe the leader's output and then decides its own output to maximize its profit.

- The follower's reaction function shows the optimal output of the follower for any given output of the leader.

(3) Equilibrium Output:

- The equilibrium is reached when the leader's chosen output and the follower's reaction to it are both optimal.

The leader's output choice takes into account the follower's best response, leading to a "Stackelberg Equilibrium."

Mathematical Representation

(1) Leader's Profit Maximization:

- Suppose there are two firms, Firm L (Leader) & Firm F (Follower).

The leader's profit function is given by:

$$\pi_L = P(Q) \cdot q_L - C_L(q_L)$$

- where $P(Q)$ is the market price as a function of total output Q (i.e., $Q = q_L + q_F$), q_L is the leader's output, and C_L is the cost function of the leader.

(2) Follower's Reaction Function:

- The follower's Reaction Function $R_F(q_L)$ is derived from its output profit maximization problem:

$$q_F = R_F(q_L)$$

- where q_F is the follower's output based on the leader's output q_L .

(3) Stackleberg Equilibrium:

- The leader sets its output q_L by maximizing its profits, considering the follower's reaction:

$$q_L^* = \arg \max [P(q_L + R_F(q_L)) \cdot q_L - C_L(q_L)]$$

- The follower then produces:

$$q_F^* = R_F(q_L^*)$$

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Conclusion:

The Stackelberg model highlights how the strategic advantage of moving first (as the leader) allows a firm to influence the market outcome in its favor, while the follower adapts to the leader's decisions.

This model provides insight into competitive strategies in market where firms have different levels of influence.

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